

Where does delivery time build up?

A source-led investigation of the FlashEats order journey

THE PROBLEM TO SOLVE

FlashEats leadership reports late deliveries and unreliable ETAs. The first decision is where to investigate in the existing workflow, before choosing a technology investment. The working question is: **which order stage merits the next operational review?**

Evidence status

FACTS FROM THIS RUN

1,600 unique orders; 1,600 dispatch records;
1,495 delivered orders have complete ETA and delivery timestamps.

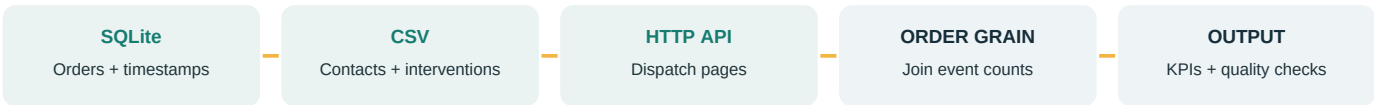
ASSUMPTIONS

Promised ETA is the commitment; actual delivery is completion; a positive difference means late. `order\_id` is stable across sources.

UNKNOWNNS

Timestamp/time-zone meaning, completeness of contacts, and stage-level service targets are not confirmed by this teaching pack.

Sources mapped to the order question



SOURCE / OWNER IN PACK	GRAIN + RETRIEVAL	WHAT IT ADDS / GAP
SQLite: orders, customers, drivers, restaurants / core operations	Entity rows; SQL	Lifecycle timestamps; repeated order IDs need a rule.
Customer interactions / support	One contact event; CSV	Contact count; not evidence that delay caused contact.
Order interventions / dispatch	One intervention; CSV	Records actions; not evidence of impact.
Dispatch / local mock API	One dispatch order; paginated HTTP	Current ETA / assignment, not historical ETA changes.

The SQLite, CSV and mock API files are supplied by the instructor's [Class 8 pack](#); the classroom inventory is in [README_STUDENTS.md](#). Results below describe this controlled teaching data, not a live business operation.

What the records say

MEASURE	RESULT	READING
Complete delivered orders	1,495 of 1,532	Late-rate denominator; 37 delivered records lack completion time.
Late delivery rate	56.39%	Actual delivery later than promised ETA on complete records.
Median order to pickup	26.07 min	Created time to pickup time.
Median pickup to delivery	47.23 min	Pickup time to actual delivery.
Orders with recorded intervention	260	At least one intervention joined to the order.

Decision supported by this evidence

Investigate the transit stage first, while checking timestamp quality. Its median duration (47 min) is longer than order-to-pickup (26 min) in this sample. This is a prioritisation signal only: without segment targets or a causal design, it does not establish that transit is the reason orders are late.

Dependability and data caveats

CHECK	EVIDENCE
Retrieval	8 API pages; 1,600 received matched the API total. One HTTP 500 and one 429 succeeded after retry.
Validation	Required fields and 30-day freshness passed. Six source rows involved in duplicate IDs; three extra rows removed. Thirty-seven delivered rows have no actual time.
Rerun behavior	Raw API pages are kept; processed outputs replace the same run-date partition; validation report and dated log are written.

Next check before an operational change

Confirm timestamp and timezone meaning with source owners; find out why completion times are missing and whether the 37 cases cluster by restaurant, driver or day; compare stage times with agreed targets. Then test one change with a comparison group. The 260 intervention records are observational and do not show impact.

Project evidence: `data/processed/run\_date=2026-09-27/metrics.json`, `order\_journey.csv`, `validation\_report.json`; pipeline: `run\_pipeline.py`. Assumed delivered-status denominator; cancelled orders are excluded. Data and starter pipeline credited to the instructor-provided repositories above.